TAMBAN THESIS CLAIM AUDIT

Statement-by-Statement Review Against the Current Repository, Data Evidence, and Scientific Validation Status

Audit date: 6 October 2026
Manuscript basis: current June 2026 thesis draft
Project basis: fresh October 2026 repository audit and generated analysis evidence

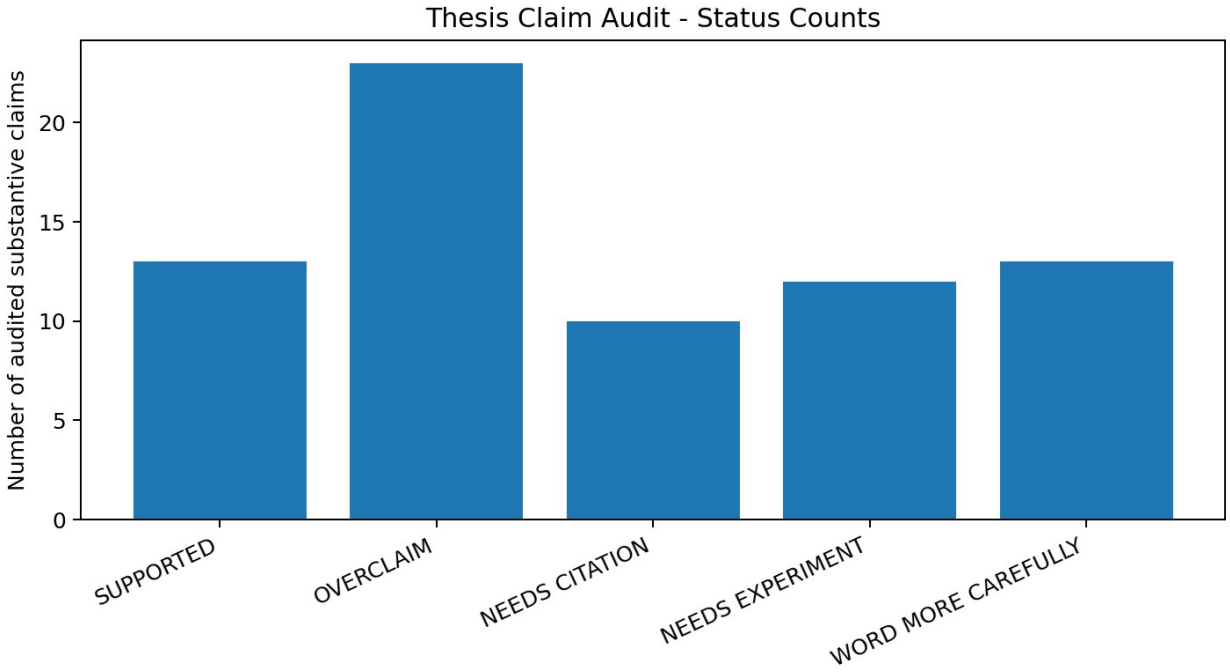
Executive Summary

Primary conclusion: the manuscript is materially out of sync with the system that now exists. The draft is centered on a CIELAB/texture regression engine that predicts a continuous Desiccation Index, while the current repository is a two-model computer-vision system for whole-fish detection, grade-aware Head/Body/Tail detection, association/tracking, and evidence-based A/B/C/Rejected/Ungraded grading. This mismatch is the single largest submission risk.

Audit fact	Current evidence
Current architecture	Model 1 whole-fish detector + Model 2 12-class grade-part detector + crop/full-frame association + grading policy.
Current grading policy	Body 50%, Head 30%, Tail 20%; missing evidence is no longer renormalized. Ungraded is distinct from Rejected.
Fresh test run	277 passed, 1 failed, 3 skipped; the old four-failure state is largely resolved.
Packaged source dataset	910 source images, 9,723 part annotations, 12 grade-part classes; raw split 861 train / 32 valid / 17 test.
Scientific limitation	Specimen-independent final validation is not yet certified; current validation metrics are development evidence, not final thesis accuracy.
Handcrafted features	Color/texture/morphology features exist, but most do not directly change the default final grade.
Current Model 2 saved validation	Precision 0.8454, Recall 0.8297, mAP50 0.8147, mAP50-95 0.5048; development/compatibility evidence only.

Claim status distribution

SUPPORTED: 13, OVERCLAIM: 23, NEEDS CITATION: 10, NEEDS EXPERIMENT: 12, WORD MORE CAREFULLY: 13



How to Read the Audit

Status	Meaning
SUPPORTED	The current code/data/evidence supports the claim as written or with only minor factual updates.
OVERCLAIM	The manuscript states or implies a capability/result that the current system/evidence does not support.
NEEDS CITATION	The claim may be reasonable, but it needs an authoritative external source or a stronger literature-search record.
NEEDS EXPERIMENT	The claim is an empirical hypothesis and requires project-specific testing before it can be stated as a finding.
WORD MORE CAREFULLY	The underlying idea may be valid, but the wording is too broad, ambiguous, outdated, or scientifically imprecise.

Top Submission Risks

1. **1. Title-method mismatch:** The title, problem statement, objectives, scope, methodology, and evaluation all describe a regression/desiccation-index thesis that is not the current system.
2. **2. Desiccation/moisture overclaim:** The current project does not establish laboratory moisture content or a validated continuous desiccation target.
3. **3. Handcrafted feature overclaim:** CIELAB/GLCM/LBP/morphology exist as research features, but most are not in the default final-grade decision path.

4. **4. Evaluation mismatch:** MAPE/RMSE/R^2 are listed, but the current system needs detection, association, fish-level classification, abstention, and tracking metrics.
5. **5. Ground-truth mismatch:** The draft says ground truth combines repeated weight-loss measurements with expert assessment; current trained assets are grade-part annotations, and specimen independence remains incomplete.
6. **6. Reference integrity:** At least one bibliography item is materially misattributed, and several others should be verified against original sources before submission.

Front Matter & Title

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
8, 37	The study is an "Analytical AI Regression Engine Using CIELAB Chromatic Moments and Geometric Texture Descriptors."	**OVERCLAIM**	The current repository is no longer primarily a regression engine. The active thesis system uses Model 1 whole-fish detection, Model 2 12-class Head/Body/Tail detection, tracking/association, and rule-based evidence fusion. Handcrafted features are mostly research/supporting signals and do not drive the default final grade.	Change the title to describe the actual two-model detection/part-grading architecture unless the regression engine is restored and becomes the real evaluated system.	Critical
8	The system performs automated desiccation grading.	**WORD MORE CAREFULLY**	The current implementation grades A/B/C/Rejected/Ungraded from visual part evidence. It does not currently estimate laboratory-validated moisture content or a validated continuous desiccation variable.	Use "quality grading" or "visual quality grading" unless expert labels and an independently validated desiccation target establish that A/B/C truly represent desiccation.	Critical
60-67	Abstract is still a template placeholder.	**SUPPORTED**	The manuscript contains instructions for writing the abstract rather than the actual abstract.	Write the abstract only after final test results are locked; include actual dataset, models, end-to-end metrics, limitations, and no unverified claims.	High

Line numbers refer to the current 700-line manuscript text used for this audit.

Chapter I - Background

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
134	Dried fish is widely consumed in the Philippines because of affordability, shelf life, and cultural significance.	**NEEDS CITATION**	Plausible contextual claim but it is an external factual statement and should be supported by BFAR/PSA/FAO or peer-reviewed literature.	Add a current authoritative source; avoid "most widely consumed" unless the source explicitly supports that ranking.	Medium

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
134	Tamban is an important protein source for many Filipino households.	NEEDS CITATION	Nutritional and consumption prevalence claims require evidence.	Cite an authoritative nutrition/fisheries source or soften to "is consumed as a protein-rich fish product."	Medium
134	Traditional drying commonly uses direct sunlight and drying readiness relies largely on visual inspection/personal experience.	NEEDS CITATION	This is central to the research problem and should be supported by fish-processing literature and, ideally, local expert evidence.	Support with fish-drying literature plus your annotator/expert protocol.	High
136	Under-drying reduces shelf life/increases spoilage risk and over-drying causes undesirable texture, reduced quality, and economic loss.	NEEDS CITATION	These are food-science consequences, not repository findings.	Cite drying/food safety studies; separate microbial safety from market-quality effects.	High
138	Computer vision can provide objective and repeatable food-quality evaluations while minimizing human bias.	WORD MORE CAREFULLY	Repeatability can improve under controlled conditions, but AI can reproduce dataset/annotation bias. Your current background-generalization and specimen-independence risks show that "minimizing human bias" is too broad.	Say computer vision "can support standardized and repeatable visual assessment under validated operating conditions."	High
140	CIELAB + chromatic moments + texture descriptors can estimate desiccation levels through regression.	NEEDS EXPERIMENT	This is an empirical hypothesis. The current active system does not use these handcrafted features as the final grade engine, and no validated continuous desiccation regression result is present.	Frame as a prior/proposed feature-engineering hypothesis or remove from the core system description.	Critical
142	Few studies focus on automated dried-sardine desiccation grading and public datasets are scarce.	NEEDS CITATION	A research-gap claim requires a documented literature search, not assertion.	Document databases, keywords, date range, and the studies found; word as "the reviewed literature identified..." rather than universal absence.	High
144	The proposed system estimates desiccation levels and gives drying-readiness recommendations and grade predictions.	OVERCLAIM	The current repository performs visual detection and part-based grade evidence; it does not currently output a validated continuous desiccation index from weight-	Rewrite around individual-fish detection, part-based evidence, A/B/C/Rejected/Ungraded decisions, and uncertainty handling.	Critical

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
			loss ground truth.		
144	The system supports more objective, consistent, and accessible drying assessment.	NEEDS EXPERIMENT	Objectivity/consistency/ accessibility are outcomes requiring user/system evaluation. Current internal-development detector metrics do not prove these benefits.	Use future-tense goals until end-to-end and user evaluation support the claim.	High

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Chapter I - Problem & Objectives

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
147	Subjective human observation may cause inconsistent grading decisions.	NEEDS CITATION	Reasonable problem statement, but should be supported by literature and preferably inter-rater/expert evidence.	Cite literature and, if possible, quantify expert agreement on a small reference set.	High
153-159	The thesis research questions are centered on a CIELAB/texture regression engine and desiccation index.	OVERCLAIM	These questions no longer match the implemented repository or current experiments.	Replace with questions on Model 1 detection, Model 2 part grading, part-to-fish association, evidence fusion, false positives/generalization, and end-to-end fish-level performance.	Critical
164-173	The objectives are to develop/compare regression models, generate a Desiccation Index, and evaluate regression metrics.	OVERCLAIM	The actual development path is two-stage object detection + part evidence fusion. Current Model 2 experiments compare detector architectures, not SVR/RFR/PLSR desiccation regressors.	Rewrite objectives to match the current system and actual measured outputs.	Critical
173	The project develops a web interface for image-based assessment.	SUPPORTED	The repository includes a web/API application path and frontend outputs.	Keep, but describe the actual stack used (FastAPI/static frontend unless the implementation changes).	Low

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Chapter I - Scope & Limitations

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
176-191	The scope is a regression-based desiccation system using CIELAB moments and geometric texture descriptors.	OVERCLAIM	This is an outdated architecture description relative to the current repository.	Replace with current Model 1/Model 2/association/fusion scope; move handcrafted features to supporting/research analysis unless validated for fusion.	Critical
187	The study is limited to Tamban/sardines.	SUPPORTED	The repository and datasets are specifically built around dried sardine/tamban grading.	Keep and clarify the exact biological species if verified; avoid implying all Sardinella species are interchangeable.	Low
188-190	Results apply to the defined capture protocol, observed drying conditions, and visual characteristics.	SUPPORTED	This is an appropriate limitation and is consistent with observed background/lighting sensitivity.	Keep and expand with camera/background/lighting/generalization boundaries.	Low
193	The system does not measure moisture directly but estimates desiccation from visual patterns and drying references.	OVERCLAIM	The first half is supported; the second half is not supported by the current active grading system because a validated desiccation target/regression is absent.	Use: "The system does not measure moisture content; it assigns visual quality grades from image-based detection evidence."	Critical

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Chapter I - Significance

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
198	The tool provides objective evaluation and may support more consistent decisions.	WORD MORE CAREFULLY	"May support" is appropriately cautious, but "objective" should be qualified by validation conditions and ground-truth quality.	Say "computer-assisted, standardized visual assessment" until independent testing and expert comparison are complete.	Medium
201	The system is a low-cost alternative to laboratory-based moisture assessment.	OVERCLAIM	The system does not directly estimate validated moisture content and therefore should not be presented as an alternative to laboratory moisture testing.	Say it may complement visual inspection; explicitly state it is not a substitute for laboratory moisture measurement.	Critical

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
204	The system can improve industry consistency and standardization.	NEEDS EXPERIMENT	Industry-level impact requires field validation and user/process evaluation.	Use "has the potential to support..." and reserve stronger wording for deployment studies.	Medium
210	The study demonstrates computer vision, feature engineering, regression analysis, and web technologies.	OVERCLAIM	Current thesis software strongly demonstrates computer vision and web deployment, but regression is not the active grading core; handcrafted features are mostly research-only.	Replace "regression analysis" with the actual detector/association/evidence-fusion methods unless a regression experiment becomes part of the final study.	High

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Chapter I - Definitions

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
220-241	CIELAB, chromatic moments, regression, SVR/RFR/PLSR, and Desiccation Index are core operational terms.	WORD MORE CAREFULLY	Several definitions are valid generally, but many no longer describe the active system. A definition section should reflect concepts actually used in the final methodology.	Add Model 1, Model 2, IoU, mAP, part association, evidence coverage, weighted support, Rejected, and Ungraded; remove unused regression terms unless retained experimentally.	High
238	"Tamban" is Sardinella spp.	WORD MORE CAREFULLY	Tamban is a local/common name and may refer to multiple sardine species. The dataset/species identity should be documented rather than generalized.	Use the exact species confirmed for the dataset, or state that Tamban is the local market/common name used for the sampled sardines.	Medium

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Chapter II - RRL

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
260-270	Drying reduces moisture, inhibits microbial growth,	NEEDS CITATION	These are external food-science claims.	Retain with reliable fish-drying/food-processing	Low

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
	extends shelf life, and is affected by temperature/humidity/airflow/time.			citations.	
270-280	Drying produces measurable color, texture, roughness, and morphological changes usable as indicators.	NEEDS CITATION	Plausible literature claim, but the specific relationship to dried sardines needs relevant sources.	Cite fish/drying studies and separate literature evidence from what your own dataset proves.	Medium
296	CIELAB is more effective than RGB for analyzing color changes.	WORD MORE CAREFULLY	CIELAB separates lightness/chromatic axes and is perceptually motivated, but "more effective" depends on task and must be supported empirically.	Say CIELAB provides a perceptually organized representation useful for color analysis; do not assert superiority without experiment.	Medium
298	CIELAB is suitable for measuring drying-related variations in sardines.	NEEDS EXPERIMENT	Suitability for this dataset/classification problem needs class-separation or ablation evidence.	Keep as a hypothesis and test with B/C distributions, mutual information, permutation importance, or ablation.	High
302-311	Chromatic moments are useful for modeling sardine desiccation progression and suitable for real-time/web use.	NEEDS EXPERIMENT	General computational simplicity is plausible, but predictive value for this thesis is unproven and the active grade path does not use them.	Separate computational feasibility from predictive effectiveness; validate before claiming usefulness.	High
314-323	GLCM/LBP/texture descriptors capture drying effects and improve prediction when combined with color.	NEEDS EXPERIMENT	The repository contains texture research code, but those features are not part of the default final grade and have not been independently shown to improve end-to-end fish grading.	Use literature support plus your own ablation before saying they improve the model.	High
320	Texture is less sensitive to illumination than color.	WORD MORE CAREFULLY	Texture descriptors can still change with scale, focus, contrast, blur, and illumination. The statement is too absolute.	Say some texture descriptors may complement color under illumination changes, subject to standardized acquisition.	Medium
330-345	SVR, RFR, and PLSR are the study models used to estimate desiccation.	OVERCLAIM	These are not the active final models in the current repository.	Replace this section with the actual detector architectures and part-evidence methodology; retain regression only as historical/proposed work if truly evaluated.	Critical

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
350-355	Users can obtain automated predictions through a browser without specialized hardware/software.	WORD MORE CAREFULLY	Browser access reduces client setup, but inference still requires a machine/server capable of running the models.	Say users can access the interface through a browser while inference runs on the configured backend hardware.	Low

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Chapter II - Synthesis

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
370-390	No identified study integrates CIELAB moments, texture descriptors, regression desiccation prediction, and web deployment for dried sardines.	NEEDS CITATION	A novelty claim must be supported by a documented literature search. It is also tied to an architecture the project no longer follows.	Re-run the literature synthesis around the final two-model part-aware grading architecture and document search scope.	High
390-400	The proposed study fills the gap by estimating desiccation and drying readiness via regression.	OVERCLAIM	The present system does not establish a validated continuous desiccation estimator.	Rewrite the synthesis around automated visual quality grading and per-fish part evidence.	Critical

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Chapter III - Research Design

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
405-412	The study is developmental + experimental.	SUPPORTED	This is a reasonable high-level research design for building a system and experimentally evaluating models/policies.	Keep, but redefine the experimental component around detector comparison, association, fusion rules, and independent validation.	Low
409-412	Images are transformed into engineered features used to train regression models that output Desiccation Index.	OVERCLAIM	Current active grading is not this pipeline.	Replace with: image -> Model 1 parent detections -> Model 2 grade-part detections -> association/crop strategy -> evidence fusion -> A/B/C/Rejected/Ungraded.	Critical

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Chapter III - Data Source

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
417-426	Primary data are drying-stage images captured by smartphones/ESP32-CAM at intervals; target 1,000-2,000 images.	NEEDS EXPERIMENT	This reads as a planned protocol, but the current packaged dataset is 910 source images with 9,723 part annotations and does not by itself prove the stated longitudinal drying-stage capture design.	Describe the dataset actually used, then separately describe any new capture protocol planned for final specimen-independent validation.	High
426	The dataset spans multiple sessions and environmental conditions.	NEEDS EXPERIMENT	This must be demonstrated from capture metadata, not assumed. Current specimen/session metadata remain incomplete for final independence certification.	Report actual session/background/lighting counts and how they are split.	High

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Chapter III - Instruments

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
443	ESP32-CAM and smartphone cameras are the image-acquisition devices used.	NEEDS EXPERIMENT	Only include devices actually used in the final dataset collection. The repository audit alone does not establish that both devices generated the training/evaluation data.	List actual hardware models/settings and associate each image/session with camera metadata.	Medium
450-464	Temperature, humidity, weather, drying duration, batch ID, drying stage, readiness, and grade will be recorded.	WORD MORE CAREFULLY	This is a good protocol, but it should be stated as implemented only if those records exist and are used/archived.	Convert to a finalized capture manifest and report completeness; do not imply variables exist if they were not recorded.	Medium

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Chapter III - Procedure

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
479-503	Fresh Tamban are dried on racks, images captured at fixed intervals with fixed camera position/distance, and experts validate stages/thresholds.	NEEDS EXPERIMENT	This is a methodology commitment that needs actual procedural logs and expert documentation.	Keep only if this protocol is actually executed and archived; add specimen IDs and keep each physical fish in one split.	High

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Chapter III - Ground Truth

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
505-508	Ground truth is created by combining repeated weight-loss measurements and expert evaluations into Desiccation Index labels.	OVERCLAIM	Current training/evaluation assets do not use a validated continuous ground-truth Desiccation Index. Current classes are annotated grade-part detections.	Replace with actual annotation ground truth: per-image/per-part labels, parent fish IDs if available, visibility/association state, and expert grade definition.	Critical
518-520	Percent weight loss is a Desiccation Index and higher values mean lower moisture content.	WORD MORE CAREFULLY	Percent mass loss is not equivalent to moisture content unless initial dry matter and other mass changes are accounted for. Calling it a Desiccation Index may be acceptable as an operational index, but not as direct moisture percentage.	Call it "percentage mass loss" unless experimentally calibrated against moisture content; validate any mapping to readiness/grade.	Critical

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Chapter III - Data Analysis

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
523-529	All images undergo resizing, noise reduction, brightness normalization, and color correction.	OVERCLAIM	The current repository has active model preprocessing and feature code, but the manuscript lists a generic	Document exact preprocessing separately for Model 1, Model 2, and research features. Do not list steps that are not	High

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
			preprocessing sequence that is not demonstrated as the single runtime/training pipeline.	actually executed.	
531-550	The final study extracts CIELAB moments and texture features (contrast, correlation, energy, homogeneity, entropy) as its two major feature sets.	WORD MORE CAREFULLY	These research features exist conceptually/code-wise, but the fresh audit shows handcrafted color/texture/morphology features do not directly affect the default final grade. Model 2 learned features dominate the active grading path.	State that these are exploratory/supporting engineered features unless ablation shows they improve end-to-end grading.	Critical
550	These texture descriptors capture drying progression.	NEEDS EXPERIMENT	The descriptor definitions are valid, but linking them to drying progression in this dataset requires empirical evidence.	Demonstrate class/severity trends and independent predictive value before making this claim.	High

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Chapter III - Model Development

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
551-558	The study evaluates SVR, RFR, and PLSR.	OVERCLAIM	Current final model development instead centers on object detectors; the latest Model 2 comparison used YOLO26m, Faster R-CNN V2, and an EfficientNetV2-M RetinaNet configuration.	Replace the regression comparison with the actual architecture comparison and clearly label internal-development vs final-test evidence.	Critical

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Chapter III - Grade Mapping

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
565-580	Desiccation Index 0-40=C, 41-75=B, 76-100=A.	OVERCLAIM	These thresholds are preliminary and are not the grading logic used by the	Remove this mapping unless expert consultation + validation specifically	Critical

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
			current repository. The active system uses grade-aware part detections with Body/Head/Tail weights and an Ungraded state.	establishes it. Describe the current 50/30/20 evidence policy instead.	
580	Thresholds will be finalized through expert validation.	SUPPORTED	This is a sound methodological safeguard and aligns with the newly created annotator/expert protocol.	Keep, but add a second empirical validation step: compare expert-informed weights/thresholds against alternatives on validation data before locking them.	Medium

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Chapter III - System

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
589-603	The system contains an AI Regression Module and Grade Mapping Module based on Desiccation Index.	OVERCLAIM	Current runtime is detector/association/grading-policy based, not regression-index based.	Replace with Model 1 detector, tracking/parent identity, Model 2 part detector, association/crop handling, grading policy, API/frontend.	Critical

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Chapter III - Tools

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
608-627	Backend = Flask/Django; frontend = React; database = MySQL.	OVERCLAIM	The audited current repository uses FastAPI and a static frontend path; the listed stack is not an accurate description of the implemented system.	Document only the stack actually used.	High
608-627	Python/OpenCV/NumPy/ Pandas/GitHub/VS Code are development tools.	SUPPORTED	These are consistent with the repository/tooling.	Keep, but separate runtime dependencies from research-only packages.	Low

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Chapter III - Evaluation

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
632-636	Functional testing verifies upload, feature extraction, prediction, and result display.	SUPPORTED	The repository has a substantial automated test suite and web/API functionality, though tests also cover more than these four items.	Expand to include model loading, class-map checks, grading semantics, association ambiguity, and failure handling.	Low
638-645	Model performance is evaluated with MAPE, RMSE, and R^2.	OVERCLAIM	These are regression metrics and do not match the current detector/classification system. Current model evidence uses precision, recall, AP/mAP and end-to-end grade metrics should use confusion matrix, macro-F1, balanced accuracy, etc.	Replace with layer-specific metrics for Model 1, Model 2, association, final grade, Ungraded coverage, and video tracking/counting.	Critical
647	Fish processors will assess usability/usefulness.	NEEDS EXPERIMENT	This is a planned UAT claim; keep future tense until executed.	Define participants, instrument, task scenarios, acceptance criteria, and analysis before collecting data.	Medium
649	Cronbach alpha will determine instrument reliability.	WORD MORE CAREFULLY	Cronbach alpha is meaningful only for a multi-item scale intended to measure a coherent construct; it is not a generic reliability stamp for all survey items.	Use alpha only for appropriate grouped Likert items, report item structure/sample size, and consider content validity/expert review separately.	Medium

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Current System - Replacement Claims

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
Repository audit	Model 1 detects/localizes whole fish and supports tracking/counting.	SUPPORTED	This matches the current repository architecture and test/audit evidence.	Use as a core methodology claim, with exact checkpoint/version and independent validation metrics when final.	Low
Repository audit	Model 2 is a 12-class detector	SUPPORTED	The class mapping is explicit in	Use this wording; do not call	Low

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
	for A/B/C/Rejected × Body/Head/Tail.		the current repository and evaluation artifacts.	Model 2 a whole-fish classifier.	
Repository audit	Final grade uses Body 50%, Head 30%, Tail 20% and preserves missing evidence rather than renormalizing it.	**SUPPORTED**	Fresh audit verified the old renormalization defect is fixed.	Use, but describe 50/30/20 as expert-informed/provisional until the annotator consultation and ablation validate it.	High
Repository audit	Ungraded is distinct from Rejected.	**SUPPORTED**	Current grading semantics support insufficient evidence/ambiguity as Ungraded rather than equating nondetection with a defect.	Make this distinction explicit in Methodology, Results, UI, and defense.	Low
Repository audit	Full-frame association exists, but crop mode remains the default runtime.	**SUPPORTED**	Fresh audit classified crop-based integration as partially fixed.	Report crop mode as baseline and compare against full-frame association on the same overlap-heavy validation set.	High
Repository audit	Handcrafted color/texture/morphology features directly determine the final grade.	**OVERCLAIM**	Fresh feature trace shows most engineered features are research/supporting only; color has no default grade adjustment and legacy RF is not runtime.	Do not claim direct use until a validated fusion rule/model actually consumes them.	Critical
Repository audit	Current Model 2 validation metrics represent final thesis accuracy.	**OVERCLAIM**	Saved validation artifact is development evidence only; specimen independence is not certified and the final test set is not yet locked.	Label all current metrics as internal development/validation results.	Critical
Repository audit	The current repository is fully reproducible from a clean checkout.	**OVERCLAIM**	Fresh audit found one remaining failing test, missing split manifest/dependency packaging issues, and three registered runtime checkpoints absent from the ZIP.	Fix packaging/dependency/model provenance before claiming full reproducibility.	High

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References & Citation Integrity

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
653-690	Reference [2] is correctly cited as FAO 2020, Fish Drying and Smoking: Production and Quality.	OVERCLAIM	Independent bibliographic check found the known book under Peter E. Doe (1998; later Routledge reissue), not the manuscript's FAO 2020 attribution.	Verify and correct [2] before submission; do not cite a title under the wrong author/publisher/year.	Critical
653-690	Every reference in the bibliography has been verified and directly supports the in-text claim where cited.	NEEDS CITATION	Several entries look generic or difficult to verify, and some citations are methodologically indirect. The bibliography should be checked one-by-one against original/official sources.	Perform a reference integrity pass: DOI/ISBN/official URL, exact title/authors/year, and claim-to-source fit.	Critical
~667	Reference [14] (CIEDE2000) directly supports claims about CIELAB effectiveness for food drying analysis.	WORD MORE CAREFULLY	CIEDE2000 is primarily a color-difference formula paper. It is not by itself strong evidence that CIELAB predicts sardine desiccation.	Use a food-color measurement review or drying-specific paper for those claims, and cite [14] only for color-difference methodology if relevant.	High

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Chapters IV-V

Lines	Claim / statement	Status	Why this is the status	Safer wording / required action	Severity
TOC 105-108	Results and Discussion / Conclusion sections can repeat planned methodology claims before final testing.	OVERCLAIM	Current manuscript has no completed results chapter, and current internal metrics are not specimen-independent final evidence.	Populate Chapter IV only from locked experiment outputs; write Chapter V after final independent evaluation.	Critical

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Recommended Thesis Reframing

Unless the regression/desiccation-index engine is rebuilt, actually trained, and independently evaluated as the final system, the thesis should be reframed around the system that exists.

Possible title directions

- Computer Vision-Based Quality Grading of Dried Sardines Using Whole-Fish Detection and Part-Based Evidence Fusion
- Part-Aware Automated Quality Grading of Dried Sardines Using a Two-Model Computer Vision Pipeline
- Automated Dried Sardine Quality Grading Using Whole-Fish Detection, Anatomical Part Detection, and Evidence-Based Decision Fusion

Use "desiccation" in the title only if the expert-defined grades and final ground truth genuinely represent desiccation/drying state rather than general visual quality.

Safer Current Problem Statement

How can a computer-vision system detect individual dried sardines, associate grade-aware Head/Body/Tail evidence with the correct fish, and produce reliable A/B/C/Rejected/Ungraded outcomes under multiple-fish, overlap, occlusion, background, and uncertainty conditions?

Suggested Specific Research Questions

7. How accurately can Model 1 localize and count individual fish while limiting false positives across representative backgrounds?
8. How accurately can Model 2 detect the 12 grade-part classes (A/B/C/Rejected × Body/Head/Tail), especially B versus C and tail classes?
9. Which integration strategy performs better end-to-end: Model 1 crop → Model 2, or full-frame Model 2 with explicit part-to-parent association?
10. How should Head/Body/Tail evidence be weighted and gated so that missing or ambiguous evidence results in Ungraded rather than a forced class?
11. Do engineered color, reflectance, texture, and geometry features improve fish-level grading beyond Model 2 evidence alone?
12. How well does the locked system generalize to new physical fish, mixed-grade scenes, overlap/occlusion, and representative acquisition conditions?

Claims You Can Safely Make Now

- The repository implements a two-model computer-vision architecture in which Model 1 localizes parent fish and Model 2 detects grade-aware anatomical regions.
- Model 2 uses 12 part classes corresponding to A/B/C/Rejected across Body/Head/Tail.
- The current grading policy uses original Body/Head/Tail weights of 50%/30%/20% and does not renormalize missing evidence.

- Insufficient or ambiguous evidence can result in Ungraded, which is conceptually distinct from Rejected.
- Full-frame association logic exists, although crop mode remains the default runtime and should be compared experimentally.
- Handcrafted appearance/texture/morphology features exist in the repository, but most are not yet proven contributors to the default final grade.
- Current Model 2 metrics are internal development/validation evidence and should not be described as final specimen-independent thesis accuracy.
- The final study still needs specimen-aware ground truth, locked validation decisions, and an untouched independent final test.

Claims You Should Not Make Yet

- The system accurately estimates moisture content.
- The system has a validated continuous Desiccation Index.
- The system is a low-cost replacement for laboratory moisture analysis.
- CIELAB/texture features are proven to improve grading.
- 50/30/20 is scientifically optimal.
- Model 2 validation mAP is the final fish-level grading accuracy.
- The model generalizes to arbitrary backgrounds or unseen environments.
- The system can determine that a physically missing Head/Tail exists merely because the detector did not find it.
- The current system has been scientifically validated end-to-end on independent physical specimens.
- Every bibliography entry has been verified.

Bibliography Integrity Check

Critical reference issue: the manuscript cites “Fish Drying and Smoking: Production and Quality” as an FAO 2020 publication. Public bibliographic records identify the well-known work under Peter E. Doe, originally published in 1998 and reissued later. Verify the exact source you actually used and correct the author/publisher/year before submission.

Reference	Audit status	Action
[2] Fish Drying and Smoking: Production and Quality	VERIFY / CORRECT	Known bibliographic records point to Peter E. Doe (1998; later reissue), not the manuscript's FAO 2020 attribution.
[14] Sharma, Wu & Dalal - CIEDE2000	KEEP ONLY FOR COLOR-DIFFERENCE METHOD	Useful for CIEDE2000/color-difference methodology, but not sufficient evidence that CIELAB predicts sardine drying/desiccation.
[1], [8], [18], [19]	VERIFY AGAINST ORIGINAL/OFFICIAL	Do not submit until exact title,

Reference	Audit status	Action
	SOURCE	author/agency, year, and retrievable official/primary source are confirmed.
General RRL citations	MATCH CLAIM TO SOURCE	A citation about generic image processing is not automatically evidence for dried-sardine desiccation, B/C grading, or your system's performance.

Defense Backup: How to Answer the Most Dangerous Claim Questions

Q: Why did your thesis title say regression when the current system is detection-based?

A: The project architecture evolved during development. Before final submission, the manuscript must be updated so the title, objectives, methodology, and evaluation describe the implemented and tested two-model system rather than the earlier regression concept.

Q: Does your model measure moisture or desiccation?

A: No direct moisture measurement is currently performed. The system is an image-based visual grading system. Any desiccation interpretation must come from expert-defined labels and independent validation, not from the detector alone.

Q: Why 50/30/20?

A: These are current design weights, not yet claimed as scientifically optimal. We are obtaining domain-expert justification and will compare the expert-informed weighting against alternatives on validation data before freezing the final policy.

Q: Are CIELAB/GLCM/texture features actually used to grade fish?

A: They are currently research/supporting features. The default final grade is driven by Model 2 anatomical grade evidence and grading policy. We will only claim an engineered feature improves grading if ablation on independent validation supports it.

Q: Is your reported mAP your final grading accuracy?

A: No. mAP evaluates object/part detection. Final fish-level grading requires separate end-to-end evaluation with per-fish ground truth, association correctness, confusion matrix, macro-F1, and Ungraded coverage.

Q: Can you say the system is validated?

A: Not yet at final thesis level. Internal development evidence exists, but specimen-independent end-to-end testing must be completed after models, thresholds, association, and grading policy are locked.

Claim-Safety Checklist Before Submission

- ☐ Change the title so it matches the final implemented architecture.
- ☐ Rewrite the Statement of the Problem and Objectives around Model 1, Model 2, association, evidence fusion, generalization, and end-to-end grading.
- ☐ Remove or clearly label the Desiccation Index/weight-loss regression concept unless it is actually implemented and validated.
- ☐ Replace the SVR/RFR/PLSR methodology with the actual model-training and comparison workflow.
- ☐ Document the actual preprocessing for Model 1, Model 2, and engineered features separately.
- ☐ Complete the annotator/domain-expert protocol before treating 50/30/20 or class definitions as domain-grounded.
- ☐ Run the weighting ablation before claiming 50/30/20 is better than alternatives.
- ☐ Create specimen IDs and keep every physical fish in one split for final validation/test.
- ☐ Lock model hashes, configs, confidence thresholds, association policy, and grading policy before the final test.
- ☐ Report current detector metrics as internal development evidence only.
- ☐ Add fish-level end-to-end metrics including A/B/C/Rejected/Ungraded, macro-F1, confusion matrix, and coverage.
- ☐ Perform overlap/occlusion and background-generalization subgroup analysis.
- ☐ Only claim handcrafted features improve grading after ablation/permutation/independent validation.
- ☐ Correct reference [2] and verify every remaining bibliography entry against original sources.
- ☐ Write Abstract, Results, Discussion, Conclusion, and Recommendations only after the final locked test is complete.

Chapter-by-Chapter Rewrite Priority

Section	Priority	Required change
Title	CRITICAL	Replace regression/desiccation-engine framing.
Abstract	CRITICAL	Write only after final results; current text is placeholder.
Chapter I Background	HIGH	Keep broad motivation; cite external claims and remove unsupported moisture/regression claims.
Problem/Objectives	CRITICAL	Rewrite to current two-model +

Section	Priority	Required change
		association + evidence-fusion research questions.
Scope/Limitations	CRITICAL	Describe actual system, operating conditions, and independent-validation limits.
Chapter II RRL	HIGH	Retain relevant CV/color/texture literature but separate literature from project evidence and rebuild synthesis around final architecture.
Chapter III Data Source	HIGH	Replace planned/assumed capture details with actual dataset provenance plus final capture protocol.
Ground Truth	CRITICAL	Describe grade-part annotations, parent/ownership/visibility labels, expert rules, specimen IDs; remove unsupported weight-loss DI ground truth.
Preprocessing/Features	HIGH	Document exact pipelines and clearly label research-only engineered features.
Training	CRITICAL	Document actual detector training/comparison and model selection.
Grading Logic	CRITICAL	Document 50/30/20, non-renormalization, evidence coverage, ambiguity, Rejected vs Ungraded, expert validation.
Evaluation	CRITICAL	Use detector + association + fish-level + coverage metrics; final untouched test only after lock.
Chapter IV	CRITICAL	Populate only with generated locked experimental results.
Chapter V	CRITICAL	Conclude only what the final test supports.

Audit Basis and Evidence Boundaries

This audit reviews substantive claims in the current 700-line manuscript draft. Administrative headings, approval-sheet wording, and empty template placeholders were not treated as scientific claims unless they create submission risk.

Project-status judgments are based on the fresh October 2026 repository audit and its generated analysis outputs. Current repository evidence includes 277 passed / 1 failed / 3 skipped tests, 910

source images, 9,723 part annotations, current grading-policy behavior, feature-flow tracing, and saved Model 2 validation artifacts.

External literature claims were not automatically accepted simply because a citation number appears in the manuscript. A claim is marked NEEDS CITATION when authoritative external support is required, and reference integrity should still be verified against original publications.

An OVERCLAIM label does not mean the idea is impossible. It means the manuscript currently states more than the implemented system or available evidence can defend.